

Energy Recovery Strategies for Batteries Providing Frequency Containment Reserve in the Nordic Power System

Jan Engelhardt^{a,*}, Andreas Thingvad^b, Jan Martin Zepter^a, Tatiana Gabderakhmanova^a and Mattia Marinelli^a

^aDepartment of Wind and Energy Systems, Technical University of Denmark (DTU), Roskilde, Denmark

^bHybrid Greentech ApS, Roskilde, Denmark

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ABSTRACT

Batteries are key components of future power systems with high shares of renewables. Their fast response enable them to contribute to one of the most crucial ancillary services - frequency containment reserve. However, longer periods with biased frequency pose challenges for batteries due to their limited energy capacity. This is particularly crucial in the Nordic synchronous area where 100 % certainty of delivery is required. The present study uses a simulation model with three years of historical frequency, market, and tariff data, to compare three energy management strategies that comply with the Nordic regulatory framework. The first two strategies both utilize an hourly adjustment of the reference power to, on average, recover the energy level to a desired value. In the first strategy, this energy is traded on the intraday market, while in the second strategy it is settled via the balancing market. The third strategy is based on an exemption agreement with the Danish transmission system operator. Since capacity payment is the strongest factor that determines the profit, the study derives maximum regulating power and annual profit as a function of the energy/power capacity ratio and the efficiency of the storage system. The highest regulating power and profit can be achieved when applying the exemption agreement. Where this agreement is not available, the battery operator should apply the second strategy and settle the energy costs through the balancing market. Trading the energy on the intraday market slightly reduces the energy costs, but allows for significantly lower regulating power and revenues.

1. Introduction

Batteries will play an essential part in future power systems due to their energy storage capability. This makes them suitable for various applications, such as electric vehicles or stationary storage systems in both industry and households [1, 2]. The fast response and bi-directional capabilities of batteries further allow these applications to offer additional value by providing ancillary grid services such as frequency regulation [3, 4]. With a decreasing number of fully controllable power sources, batteries could become a cornerstone in providing stable grid operation [5, 6, 7]. In particular, the synchronous grid of northern Europe incorporates already a high share of both renewables and battery applications [8, 9]. The Transmission System Operators (TSOs) of the Regional Group Nordic (RG-N) synchronous zone of Norway, Sweden, Finland, and eastern Denmark (DK-2), have a shared set of grid codes including the services related to frequency regulation. The focus of the present study is on frequency-controlled normal operation reserve (FCR-N), since it had the highest prices in the past years [10]. FCR-N is a symmetrical service, which requires the battery operator to offer the same power capacity for upward and downward regulation, with respect to the reference power. Fre-

quency reserves must be provided linearly according to the deviation from 50 Hz, with full power activation from deviations of ± 100 mHz [11]. Generally, the frequency pattern is normally distributed with an average frequency of 50 Hz. Consequently, the energy content associated with bi-directional frequency control is small over a longer period when neglecting conversion losses. However, when looking at shorter periods, the energy content of the frequency can be considerably biased which could fully charge or deplete units with limited energy capacity, such as batteries [12, 13]. This is particularly challenging when delivering the FCR-N service because it does not include any deadband around 50 Hz, unlike other synchronous grids like Continental Europe. Furthermore, the Danish TSO Energinet requires 100 % certainty of delivery and reserves the right to exclude battery operators from further participating in the frequency market if they repeatedly fail to deliver the service [14]. The requirement for continuous service provision has two main implications for the battery operation. First, the offered regulation capacity, in MW, must be in reasonable proportions to the nominal energy capacity of the battery in MWh. Second, the battery operation must include some strategy for managing the energy level to avoid being completely charged or depleted during longer periods of over- or underfrequency.

Energy management strategies for frequency control are already discussed in literature, however, they often make use of degrees of freedom, such as charging while the frequency is within a deadband, which are not present in the Nordic synchronous area [15, 16, 17, 18, 19, 20]. Other operating

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*Corresponding author

 janen@elektro.dtu.dk (J. Engelhardt)

ORCID(s): 0000-0003-1307-4095 (J. Engelhardt); 0000-0003-3464-4739 (A. Thingvad); 0000-0001-9387-6033 (J.M. Zepter); 0000-0003-2771-6706 (T. Gabderakhmanova); 0000-0001-5184-7269 (M. Marinelli)

Nomenclature

η	Storage efficiency	E_t	Current battery energy level
$\pi_{\text{down},t}$	Downregulation price	$E_{\text{diff},t}$	Difference between target and current energy level
$\pi_{\text{int},t}$	Intraday price	$E_{\text{ref},t}$	Target energy level of storage
$\pi_{\text{reg},t}$	Regulation price in one-price system	e_T^+	Delivered FCR-N energy for interval T
$\pi_{\text{up},t}$	Upregulation price	e_T^-	Received FCR-N energy for interval T
Ψ	Profit	f_t	Grid frequency
Y_{cap}	Capacity payment	P_{FC}	Regulating power for frequency control
ζ_{imbal}	Imbalance settlement	P_{N}	Nominal power capacity
ζ_{rec}	Costs of energy recovery	$P_{\text{ref,max}}$	Maximum value for reference power
ζ_{tariffs}	Tariff costs	$P_{\text{int},t}$	Internal power that changes stored energy
E_0	Initial storage energy	$P_{\text{ref},t}$	Reference power/operating point
E_{max}	Maximum occurred energy level	T	Time interval at electricity market
E_{min}	Minimum occurred energy level	t	Time
E_{N}	Nominal energy capacity		

strategies are based on risk management by optimizing the level of penalties for service failure in the decision making [21, 22, 23], which in RG-N could lead to an exclusion from the market. In the Nordic power system, energy management could be generally performed through an adjustment of the operating point. This refers to the reference power at a frequency of 50 Hz. Changing the reference power allows to, on average, charge or discharge the battery in order to restore the reserves. The associated energy can be either traded on the intraday market, or settled as imbalances on the balancing market (without the delay of the intraday market). The concept of adjusting the reference power was discussed for other synchronous areas that offer more favourable market structures with shorter time intervals [18, 24]. For the Nordic power system, this method has only been investigated for specific applications and storage sizes [25]. However, battery specifications such as the energy and power capacity or the system efficiency are different from one application to another. The general dependency of FCR-N power on these specifications has not been studied yet, i.e. for a given battery system it is not known what fraction of the available power capacity can be allocated for service delivery and how much should be reserved for adjusting the reference power.

Besides the strategy of controlling the reference power, the Danish TSO Energinet currently offers operators of systems with limited energy storage capacity the option to sign an exemption agreement, which applies to the condition of continuous service delivery. Instead, the agreement only requires the service to be continuously active and contain functionalities which can guarantee a 100 % delivery of power for a minimum of 15 minutes. Then, the battery operator has the chance to restore the reserves within 15 minutes. However, also when signing an exemption agreement, it is crucial to know how much of the power capacity can be used for delivering FCR-N without reaching the battery limits.

The present study addresses this issue for the three men-

tioned energy recovery strategies complying with the Danish market rules: (i) Adjusting the reference power through the intraday market, (ii) adjusting the reference power through the balancing market, (iii) recovering the energy based on an exemption agreement with Energinet. The analysis is performed with a simulation model of a battery system using three years of high resolution frequency measurements of the Nordic power system, and historical data for market prices and tariffs. For all three strategies, the study investigates how much of the total power capacity can be dedicated to providing frequency control. The performed analysis includes various factors, such as the ratio of energy and power capacity of the battery, as well as the system efficiency. Furthermore, the study highlights the expected annual profit, and estimates the number of full cycles that the battery will experience.

The remainder of the paper is structured as follows: Section 2 introduces the battery model and the different control strategies. Section 3 describes the case study and the used data. Section 4 summarizes and discusses the numerical results, followed by concluding remarks in Section 5.

2. Methodology

The methodology section is divided into three parts. First, a general overview of the model is provided. Second, three different control strategies are introduced for managing the energy level while providing FCR-N. Third, the different revenues and costs associated with FCR-N are described.

2.1. Model overview

A simulation model was developed in MATLAB & Simulink [26], allowing to test battery control strategies for providing FCR-N. Figure 1 shows a high-level diagram of the model. The physical layer of the system comprises the battery storage including its inverter, which serves as an inter-

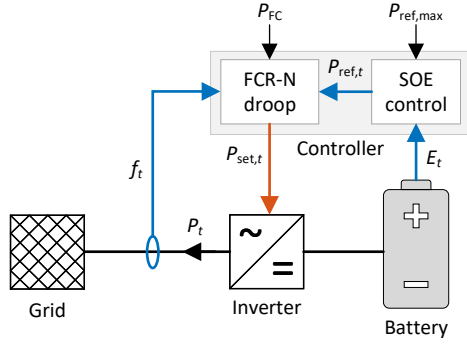


Figure 1: Overview of simulation model for testing FCR-N control strategies.

face to the grid. The inverter is used to control the battery power P_t at the grid connection point at each time instance t . Taking into account the system efficiency η , the internal power causing an increase or a reduction of the energy state in the battery is defined as:

$$P_{\text{int},t} = \begin{cases} P_t/\eta, & \text{if } P_t \geq 0 \\ P_t \cdot \eta, & \text{if } P_t < 0 \end{cases} \quad (1)$$

The power flow is defined in generator convention, with positive values corresponding to discharging, and negative values to charging. Based on the internal power, the energy level of the battery is determined as

$$E_t = E_0 - \int P_{\text{int},t} dt, \quad (2)$$

where E_0 is the energy level at the beginning of the simulation. The battery control is realized as a two-level controller. The lower control level is responsible for delivering the FCR-N service, while the higher control level is adjusting the reference power, aiming to keep the stored energy within its operational limits. In the following subsection, this control concept is described in detail.

2.2. Control strategies

FCR-N is a symmetrical service, offering the same power capacity P_{FC} for upward and downward regulation with respect to the reference power $P_{\text{ref},t}$. The reserves are provided linearly to the frequency, with full power activation from deviations of ± 100 mHz. The corresponding droop control is shown in Figure 2. For a frequency value f_t at time t , the droop characteristic is used to determine the setpoint for the battery inverter $P_{\text{set},t}$. Even though the frequency pattern is normally distributed around 50 Hz, longer periods with biased frequency can occur. This poses challenges for systems with limited energy capacity. Hence, the battery operation must include some strategy for managing the energy level to avoid being completely charged or depleted during longer periods of over- or underfrequency. This task is taken by the second control level shown in Figure 1. The block “SOE control” monitors the charged energy E_t and provides the droop reference power $P_{\text{ref},t}$. An adjustment of the value

aims at moving the charge level closer to a reference energy $E_{\text{ref},t}$ in order to ensure continuous operation. The reference power can be set to values between $P_{\text{ref},t} \in \{0; P_{\text{ref},\text{max}}\}$. The upper limit of the reference power and the maximum value of the FCR-N droop must always meet the condition

$$P_{\text{FC}} + P_{\text{ref},\text{max}} \leq P_N, \quad (3)$$

where P_N is the nominal power capacity of the storage system. Furthermore, the reference power must not be updated too frequently, as this would counteract the FCR-N service. Therefore, the control proposed in the present paper schedules an hourly change in the reference power, to align with the time intervals of the Nordic intraday market. All in all, three different control strategies are introduced and compared in the following.

Energy management through the intraday market

The first strategy involves an hourly adjustment of the reference power, where the associated energy is traded on the intraday market. Bids for the intraday market have to be made at least one hour before the start of the respective hour. Consequently, the actual activation of the reference power adjustment occurs in 1-hour delays. The aim of the adjustment is to move the energy level E_t to the target $E_{\text{ref},t}$, decreasing the difference $E_{\text{diff},t} = E_t - E_{\text{ref},t}$. As demonstrated by Zhang et al [21], the target $E_{\text{ref},t}$ is a single value (rather than a deadband) in systems where the service failure penalty is much larger than the electricity price. Since in the Nordic regulatory framework service failure can lead to being excluded from the market, this method of charge level control was applied in the present study. In the intraday market, the desired reference power during time interval T_i (starting at time t_i) is determined one hour before at time $t_i - 1$ h through

$$P_{\text{ref},T_i} = \begin{cases} E_{\text{diff},t_i-1\text{h}}/(1\text{h}) \cdot \eta, & \text{if } E_{\text{diff},t_i-1\text{h}} \geq 0 \\ E_{\text{diff},t_i-1\text{h}}/(1\text{h})/\eta, & \text{if } E_{\text{diff},t_i-1\text{h}} < 0, \end{cases} \quad (4)$$

taking into account the expected power losses by including the storage efficiency η . The calculation of revenues and costs is constituted in section 2.3.

Energy management through the balancing market

The second strategy also involves an hourly adjustment of the reference power, but settles the associated energy as imbalances through the balancing market. The energy cost is unknown as the imbalance price is settled post operation and might be higher compared to the first strategy. Yet, settling the energy as imbalances removes the 1-hour delay of the intraday market, allowing a more direct control of the charge level. The desired reference power during time interval T_i is determined at time t_i (start of interval) through

$$P_{\text{ref},T_i} = \begin{cases} E_{\text{diff},t_i}/(1\text{h}) \cdot \eta, & \text{if } E_{\text{diff},t_i} \geq 0 \\ E_{\text{diff},t_i}/(1\text{h})/\eta, & \text{if } E_{\text{diff},t_i} < 0. \end{cases} \quad (5)$$

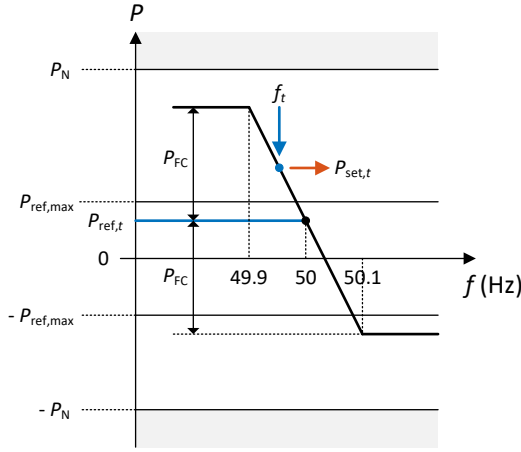


Figure 2: Droop control for FCR-N provision.

Exemption agreement

The exemption agreement offered by the Danish TSO Energinet relaxes the requirement for continuous service delivery. It specifies that the service must only “be continuously active and contain functionalities which can guarantee a 100 % delivery of power for a minimum of 15 minutes”. Then, the battery operator has the chance to restore reserves within 15 minutes. In practice, the power setpoint defined by the FCR-N droop is integrated to estimate the already delivered energy. If this energy value exceeds $\pm P_{FC} \cdot 15 \text{ min}$, the battery will use the following 15 minutes to recover its charge level E_t to $E_{ref,t}$. In the model, the droop reference power is set to $P_{ref,t} = 0 \quad \forall t$.

2.3. Profit

This subsection provides a breakdown of how the revenues and costs for FCR-N provision are constituted. For each profit component, differences between the three control strategies are highlighted.

2.3.1. Capacity payment

Capacity payment is the main revenue stream for FCR-N and is paid per available power capacity, $P_{FC,T}$, for each hourly time interval T . The service is traded on market terms at one and two days ahead. Service providers submit bids for their hourly power capacities with respective prices $\pi_{cap,T}$. All bids are sorted according to the price per MW in ascending order and selected to cover Energinet’s combined required volume with the least costs. Each accepted bid is settled with the pay-as-bid mechanism. Considering successful market participation, the provider’s revenues through capacity payment for various time intervals T are calculated as

$$Y_{cap} = \sum_T P_{FC,T} \cdot \pi_{cap,T}. \quad (6)$$

In the present study, we assume that the battery operator provides continuous operation with the same power capacity for all time intervals.

2.3.2. Imbalance settlement

The provision of FCR-N involves an electric energy uptake or delivery by the battery, as over- and underfrequency events occur. The cost settlement for this energy imbalance is calculated by the TSO by separately integrating the positive and negative power response for each hour T , resulting in an estimation of the delivered energy e_T^+ and the received energy e_T^- , defined in per unit. The calculated values are based on the measured frequency assuming an ideal response from the unit. The total delivered and received energy are determined by multiplying with the FCR-N power capacity $P_{FC,T}$. The delivered energy is paid according to the up-regulation price $\pi_{up,T}$, while the received energy is charged according to the down-regulation price $\pi_{down,T}$. Consequently, the settlement for the energy imbalance for several time intervals T is calculated as

$$\zeta_{imbal} = \sum_T P_{FC,T} \cdot (e_T^+ \cdot \pi_{up,T} + e_T^- \cdot \pi_{down,T}). \quad (7)$$

Since the up-regulation price is generally greater than or equal to the down-regulation price, the compensation scheme for the exchanged energy is expected to result in positive revenues for the service provider.

2.3.3. Energy recovery

The costs for recovering the energy level of the battery depend on which of three control strategies is applied. When following the first strategy (intraday market), the energy $P_{ref,T} \cdot \Delta T$ connected to the reference power for the length of market interval T must be traded at least one hour before on the intraday market. In this case, the energy recovery costs are given as

$$\zeta_{rec,T} = \pi_{int,T} \cdot P_{ref,T} \cdot \Delta T, \quad (8)$$

where $\pi_{int,T}$ is the intraday energy price for the respective market interval.

When applying the second strategy (balancing market), the energy associated with the reference power adjustment, $P_{ref,T} \cdot \Delta T$, is settled through the balancing market. In contrast to the imbalance settlement for FCR-N provision, these costs are based on the regulating prices using the one-price system, since batteries are considered as consuming units:

$$\zeta_{rec,T} = \pi_{reg,T} \cdot P_{ref,T} \cdot \Delta T. \quad (9)$$

The same scheme is also applied to the third strategy, where an exemption agreement is made with Energinet. Hence, the price for the energy exchanged with the grid during a 15-minute recovery period is also calculated using equation (9). For all three control strategies, the overall costs of recovering the SOC are calculated as the sum of the costs per recovery action:

$$\zeta_{rec} = \sum_T \zeta_{rec,T}. \quad (10)$$

2.3.4. Tariffs

When delivering FCR-N, tariffs have to be paid for the energy that was exchanged with the grid, based on the power P_t at the grid connection point. The tariffs include payments to the TSO and the DSO. Historical TSO tariffs are published by Energinet [27]. For imported energy, they include system tariffs (“Systemtarif”), grid tariffs (“Nettarif”), and balancing tariffs for consumption (“Balancetarif for forbrug”), and for exported energy feed-in tariffs (“Indfødningsstarif”) and balancing tariffs for production (“Balancetarif for produktion”). DSO tariffs arise only for imported energy and are dependent on the time of the day, the day of the week, and the season. The tariff schemes used in the present study were taken from the Danish DSO *Radius* [28].

The total profit of providing FCR-N, excluding investment costs, is calculated by adding the values for capacity payment, imbalance settlement, energy recovery, and tariffs:

$$\Psi = \Upsilon_{\text{cap}} + \zeta_{\text{imbal}} + \zeta_{\text{rec}} + \zeta_{\text{tariffs}}. \quad (11)$$

3. Case study

The case study is structured in two parts. In the first part, the data used as a simulation input is specified. The second part describes the design of the study.

3.1. Input data

The simulations are performed using three years of historical data from 2018–2020. The grid frequency time series is based on measurements of the Nordic power system provided by the Finnish TSO Fingrid [29]. The data is resampled to achieve a constant time resolution of 10 s, and gaps in the measurements (1.48 % of the sample points) are filled with values of 50 Hz. Market data for intraday, day-ahead, and balancing market for DK-2 are obtained from the Nord Pool website [30] and from the data service of Energinet [31]. For the time series of intraday prices, the average market price of each hour is used. Prices for FCR-N service are also considered with the average market price of each hour, as provided by Energinet [32]. TSO tariffs were retrieved from Energinet [27], and DSO tariffs for customer type “B-høj” (10 kV grid connection) from Radius [28].

3.2. Design of the study

The study aims at assessing the performance of the three control strategies introduced in section 2.2. For each of the strategies, two main questions are addressed: (i) which fraction P_{FC}/P_N of the nominal power can be dedicated to delivering FCR-N, and (ii), what is the expected annual profit of the service. Both questions depend on various factors, such as the battery capacity and the overall efficiency of the storage system. Furthermore, the first two control strategies have a second main control parameter, the fraction of the nominal power dedicated to adjusting the reference power. Therefore, the following procedure was defined. The nominal storage power P_N is initially set to a fixed value of 1 MW.

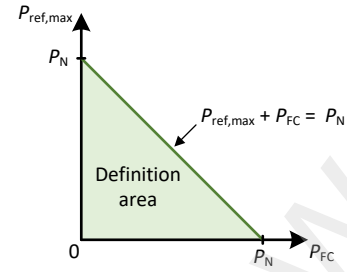


Figure 3: Definition area for droop and reference power.

Then, a set of simulations is performed, where between each new run the values of P_{FC} and $P_{\text{ref,max}}$ are varied in steps of 50 kW to cover the complete definition area shown in Figure 3. The definition area has three borders, namely, $P_{\text{FC}} = 0$ kW, $P_{\text{ref,max}} = 0$ kW, and $P_{\text{FC}} + P_{\text{ref,max}} = P_N$. During each run, the progression of the stored energy E_t is observed, without defining any limits of the energy capacity. Both the initial value E_0 and the target value E_{ref} are set to 0 kWh. Finally, the required energy capacity of the storage is determined as

$$E_N = 2 \cdot \max(|E_{\min}|, E_{\max}), \quad (12)$$

where $E_{\min}$ is the minimum, and $E_{\max}$ is the maximum value of E_t that occurred during the complete simulation period of three years. Since each combination of P_{FC} and $P_{\text{ref,max}}$ results in a required energy capacity E_N , the maximum droop value P_{FC} can, conversely, be expressed as a function of the available storage capacity E_N . By dividing E_N by P_N , the results are generally applicable to any storage with given energy/power capacity ratio. For each simulation run, the achieved profit is determined and also expressed as a function of the storage size. It is noteworthy, that for the third control strategy the proposed procedure simplifies, as the reference power is set to zero. Finally, the complete procedure is conducted for different roundtrip efficiencies between 70 % and 100 %, in 5 % steps. Hence, the sensitivity of the results to the storage efficiency can be assessed.

4. Results and Discussion

The results section is structured as follows: First, the key findings for each of the three strategies are described. Second, an economic comparison between the three strategies is made for an exemplary storage size. Third, the results for the different years are compared to evaluate year-to-year variations. Finally, a sensitivity analysis is performed to assess the impact of the storage efficiency on control parameters and revenues.

4.1. Energy management through intraday market

Figure 4 shows the simulation results for strategy 1, where the energy associated to the reference power setpoint is traded on the intraday market. For each combination of the maximum droop power P_{FC} and the maximum reference power

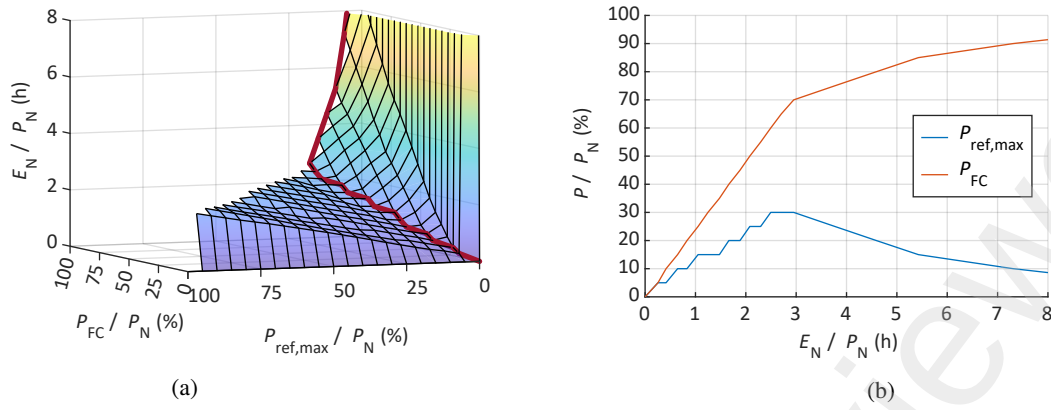


Figure 4: Simulation results for control strategy 1, energy management through intraday market; a) Required storage capacity E_N for various combinations of P_{FC} and $P_{ref,max}$, for a storage roundtrip efficiency of 90%. The red line shows the minimum required storage capacity as a function of P_{FC} ; b) Dependency of the optimal control parameters P_{FC} and $P_{ref,max}$ on the storage capacity E_N . The graph is a two-dimensional representation of the red line shown in a).

$P_{ref,max}$, the required storage capacity E_N was estimated according to equation (12). Subfigure 4a shows a 3D surface plot of the relation of P_{FC} , $P_{ref,max}$, and E_N , normalized by the power capacity P_N . When looking at the surface plot from z-direction (top view), one would perceive the triangular shape of the definition area defined in Figure 3. In the three-dimensional space, the surface has a concave shape. The progression of the minimum required energy capacity for each value of P_{FC} is indicated by the red line. Starting from the origin, the line is running through the valley of the concave surface plot until it reaches the outer edge of the definition area where $P_{FC} + P_{ref,max} = P_N$. From there, it follows along the P_N edge, in the direction of P_{FC} . The shape indicates, that for each P_{FC} there is a corresponding $P_{ref,max}$, which requires the least storage capacity. Therefore, enabling more reference power can be disadvantageous from a certain point on. This is due to the 1-hour delay between the calculation and the actual activation of the reference power, when trading the energy on the intraday market. It was observed that increasing $P_{ref,max}$ for the same P_{FC} can lead to alternating energy levels, with the reference power changing signs every hour. This is why the required storage capacity starts increasing again after reaching the valley bottom. Consequently, the red line represents the optimal combination of P_{FC} and $P_{ref,max}$ for a given storage capacity.

The progression of the curve in the three-dimensional space can be split into components of P_{FC} and $P_{ref,max}$ to achieve a two-dimensional representation, as shown in Subfigure 4b. This graph constitutes one of the key findings of the present study because it tells what fraction of the available power capacity can be dedicated to delivering FCR-N, as a function of the storage energy/power ratio. Furthermore, it provides the corresponding fraction of $P_{ref,max}$ that is required to keep the energy level within its operating limits. Until a storage capacity of around 3 h, P_{FC} and $P_{ref,max}$ increase linearly until 70 % and 30 %, respectively. At this point, the full power capacity P_N is reached, which means that for any further increase of P_{FC} , less reference power is

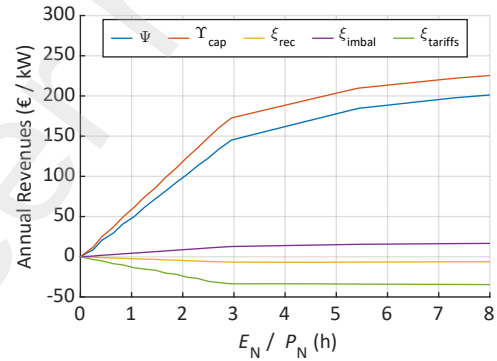


Figure 5: Annual revenues for control strategy 1, energy management through intraday market, for a storage roundtrip efficiency of 90%.

available for recovering the energy. Consequently, $P_{ref,max}$ decreases and P_N increases only with declining growth rate. For the obtained combinations of P_{FC} and $P_{ref,max}$, an economical analysis was performed, as shown in Figure 5. In the plot, the progressions of capacity payment Y_{cap} , imbalance settlement ξ_{imbal} , energy recovery ξ_{rec} , tariffs $\xi_{tariffs}$, and overall profit Ψ are shown as a function of the energy-power ratio E_N/P_N . The results show that the capacity payment is the strongest factor influencing the overall profit. Hence, the profit follows the same progression as the droop power P_{FC} derived in Figure 4b. Until a storage capacity of 3 h, the annual profit increases linearly to around 145 € per kW installed power capacity. After this point, the profit increases with declining growth rate. For a storage capacity of around 3 h, the annual profits amount to 201 € per kW. While the imbalance settlement contributes slightly to the earnings, energy recovery and tariffs reduce the overall profit. It is noteworthy, that the costs for energy recovery are insignificant compared to the capacity payment. This emphasizes once more the importance of maximizing the fraction of power capacity allocated to FCR-N provision, in order to increase the overall profit.

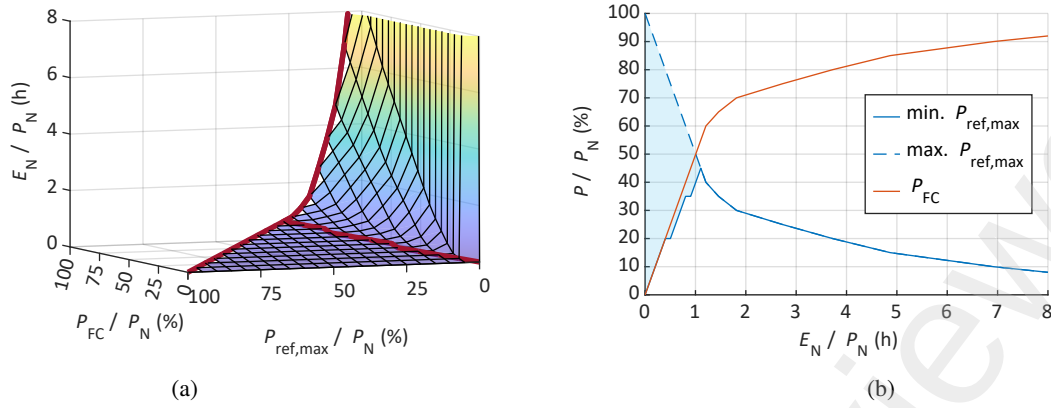


Figure 6: Simulation results for control strategy 2, energy management through balancing market; a) Required storage capacity E_N for various combinations of P_{FC} and $P_{ref,max}$, for a storage roundtrip efficiency of 90 %. The red lines mark the area with the minimum required storage capacity as a function of P_{FC} ; b) Dependency of the optimal control parameters P_{FC} and $P_{ref,max}$ on the storage capacity E_N . The graph is a two-dimensional representation of the red lines shown in a).

4.2. Energy management through balancing market

Figure 6 shows the simulation results for strategy 2, for which the energy associated with the reference power set-point change is settled through the balancing market. As for the first strategy, the required energy capacity for the different combinations of P_{FC} and $P_{ref,max}$ is presented in a surface plot, shown in Subfigure 6a. One distinctive difference is that for low values of P_{FC} , E_N does not increase again from a specific $P_{ref,max}$, but stays constant. Hence, there are several values of $P_{ref,max}$ that achieve the minimum E_N for the same P_{FC} . This is due to the fact that the control does not have the 1-hour delay of the intraday market. Consequently, increasing $P_{ref,max}$ does not lead to any disadvantage as the actually activated reference power never reaches the maximum value. To highlight this point, two lines were used that enclose all minima of E_N for each value of P_{FC} . The first line shows the minimum required $P_{ref,max}$ at which the minimum E_N is reached. The second line marks the maximum possible $P_{ref,max}$, following the edge of the definition area. Both lines merge when the first line reaches the edge of the definition area. As for strategy 1, the lines were split into components of P_{FC} and $P_{ref,max}$ to achieve a two-dimensional representation, as shown in Subfigure 6b. For the section with two lines, the area with valid values of $P_{ref,max}$ is marked in light blue. It can be observed, that the point where $P_{FC} + P_{ref,max} = P_N$ is reached for a considerably lower storage capacity of 1.1 h, compared to strategy 1. At this location, droop and maximum reference power are 55 % and 45 % of the total power capacity, respectively. From there on, $P_{ref,max}$ has to decrease to enable more power capacity for FCR-N. Since less power is available for managing the energy level, the increase of P_{FC} starts to stagnate. All in all, strategy 2 allows for significantly higher droop values for the same energy capacity, compared to strategy 1. Furthermore, the selection of the control parameter $P_{ref,max}$ is much simpler, as it can always be set to $P_{ref,max} = P_N - P_{FC}$ for all storage sized. This makes sure that there is always enough

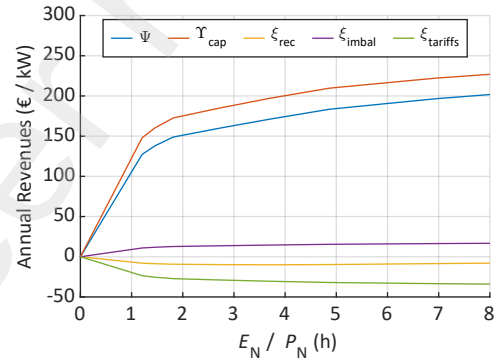


Figure 7: Annual revenues for control strategy 2, energy management through balancing market, for a storage roundtrip efficiency of 90 %.

power available for managing the energy level.

The economical results for control strategy 2 are presented in Figure 7. The trends look similar to strategy 1, with the capacity payment being the dominant factor that is driving the overall profit. However, strategy 2 achieves higher profit, in particular for low storage capacities. For higher values of E_N/P_N , the profit of strategy 2 stagnates, which allows the profits of strategy 1 to approximate. The costs for energy recovery are higher when settling the energy through the balancing market. As seen in the graph, they approximately balance out the earnings through imbalance settlement. Yet, the costs of energy recovery appear to be insignificant compared to the capacity payment.

4.3. Exemption agreement

For control strategy 3, no adjustment of the reference power is necessary, as the reserves can be recovered after an energy equivalent of 100 % full droop power for 15 minutes. Hence, the number of simulation runs decreases since only P_{FC} is varied. Figure 8 shows the simulation result for control strategy 3. The first subplot shows the maximum droop power P_{FC} as a function of the normalized storage ca-

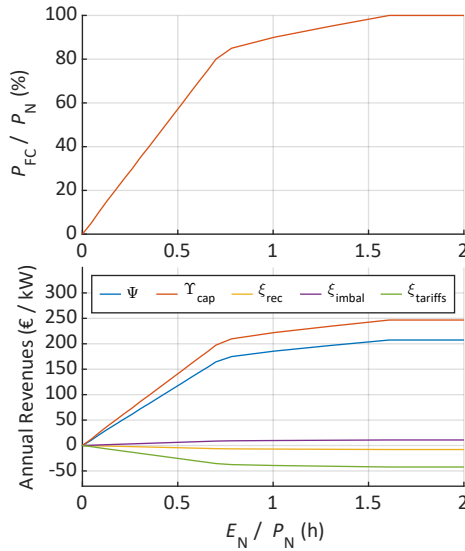


Figure 8: Maximum droop power and annual revenues for strategy 3, exemption agreement, for a storage roundtrip efficiency of 90 %. Note the different x-axis limits to Figure 4 and Figure 6.

capacity E_N/P_N . P_{FC} increases linearly until a value of 80 % at 0.7 h. Then, the progression stagnates and reaches 100 % at 1.6 h. All in all, the exemption allows battery operators to activate the full power capacity from significantly smaller energy/power ratios. This is also reflected in the profit, as shown in the lower subplot. As for the previous results, the capacity payment is the dominant factor that determines the overall profit. Hence, the third strategy requires much smaller storage capacities than the first two control strategies to reach an annual profit of 200 € per kW.

4.4. Economical comparison

In this subsection, an exemplary battery system is used to demonstrate how to practically apply the results presented in the previous subsection. Assuming a battery system has an energy capacity of 1.5 MWh, a nominal power of 1 MW, and a roundtrip efficiency of 90 %. Based on the ratio $E_N/P_N = 1.5$ h, both the controller parameter and the expected annual profit can be obtained for each of the three control strategies. For strategy 1, Figure 4 specifies the droop power as 35 %, and the maximum reference power to 15 % of the nominal power. This corresponds to values of $P_{FC} = 35 \% \cdot P_N = 350$ kW and $P_{ref,max} = 15 \% \cdot P_N = 150$ kW. For strategy 2, Figure 6 specifies the controller parameter as $P_{FC} = 650$ kW and $P_{ref,max} = 350$ kW. For strategy 3, Figure 8 defines the allowed droop power as $P_{FC} = 980$ kW. Furthermore, the simulation results also provide estimations for annual revenues, costs, and profit. A summary comparing the economical results of the three strategies is presented in Figure 9.

The findings emphasize large differences in capacity payment, where strategy 3 yields more than double the revenues than strategy 1. This is due to the higher power capacity for FCR-N that can be applied when following the third strategy. The higher power capacity is also leading to a higher energy

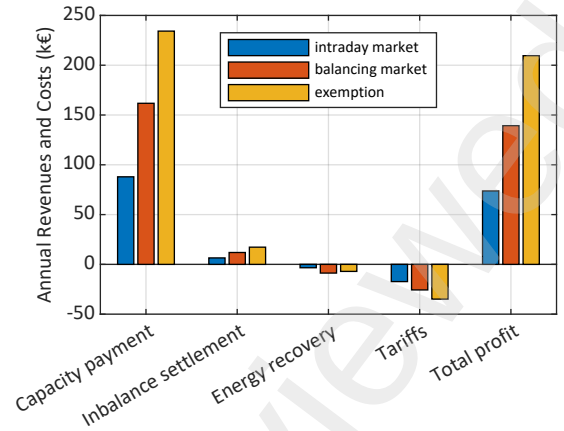


Figure 9: Economical comparison of the three control strategies. The considered battery system has an energy capacity of 1.5 MWh, a power capacity of 1 MW, and a roundtrip efficiency of 90 %.

throughput and, consequently, a higher imbalance settlement for strategy 3. Recovering the energy level through the intraday market is the cheapest option. However, the saved costs are insignificant compared to the lower revenues that are achieved when applying strategy 1. Since the tariffs are closely related to the overall energy throughput, strategy 3 has the highest tariff costs, followed by strategy 2, and then strategy 1.

All in all, the results clarify that the exemption agreement leads to significantly higher profits. In case exemption agreements are not available anymore, or for other countries than Denmark, the other two strategies have to be considered. The results clearly show that when managing the energy level through an adjustment of the reference power, the associated energy should be settled as imbalances. This allows for higher FCR-N power and higher capacity payment which is the dominant factor that determines the overall profit. For this reason, the following results and discussions focus only on strategy 2 and 3, while energy management through the intraday market is omitted to increase clarity.

4.5. Year-to-year comparison

The results presented in the previous subsections were achieved by running simulations for three consecutive years. To assess how sensitive control parameters and profits are to different periods, the simulations were also performed for the three years individually. Figure 10 shows this year-to-year comparison. The upper subplot presents the maximum droop power P_{FC} as a function of E_N/P_N for each year with control strategy 2 and 3, respectively. It can be seen that the progression of P_{FC} is robust, as the difference between the years is marginal. Specifically, when applying the exemption agreement the maximum droop power appears to be insensitive to the time period. When comparing the different progressions of strategy 2, a slightly higher droop power would have been possible in 2020. In this context, it is noteworthy that the P_{FC} progression shown in Figure 6b represents the minimum of the three lines shown in Figure 10. Hence,

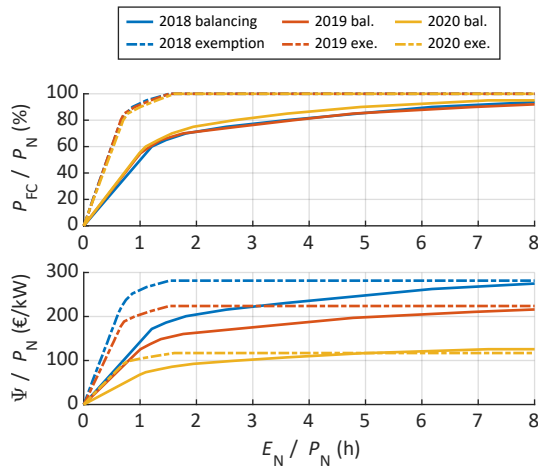


Figure 10: Year-to-year comparison of droop power and profits.

when following the recommendation provided in Figure 6b, the scheduled service could have been provided 100 % of the time for the complete three-year period.

When comparing the overall profit a few things can be noted. For both control strategies, the annual profit varies considerably from year to year. The profits in 2018 are almost three times higher than in 2020. This is mainly due to the different average prices for FCR-N regulation in the respective years. As elaborated by Thingvad [10], FCR-N prices are related to the reservoir levels of hydropower plants. For 2020, the water level was comparably high, which increases the availability of hydro power in the frequency regulation market, thus, leading to lower prices. The trend of overall decreasing prices, as Figure 10 would suggest, is refuted when looking at FCR-N prices of the last decade. Generally, regulation prices changed from year to year without clear trend. However, this should be taken into account for investment considerations, as the prices for frequency regulation for the upcoming years are both difficult to predict and vary significantly. Finally, the graph shows that the exemption agreement generally yields higher profits than strategy 2. The difference is largest for low values of E_N/P_N , and becomes smaller with increasing storage capacity.

4.6. Impact of storage efficiency

A sensitivity analysis was performed to evaluate the influence of the storage efficiency on droop power, profit, and battery cycles. One cycle is considered as a full charge and discharge of the battery, and estimated by dividing the total energy throughput by twice the nominal energy capacity. This method was chosen as some manufacturers specify warranties based on the energy throughput. The simulations were executed for the complete period from 2018 to 2020, with storage roundtrip efficiencies between 70 % and 100 %, in 5 % steps. Figure 11 summarizes the simulation results for strategy 2, where the reference power was settled through the balancing market. The first subplot shows that the progression of the maximum droop power P_{FC} shows a slight dependency on the storage efficiency. For higher efficiencies,

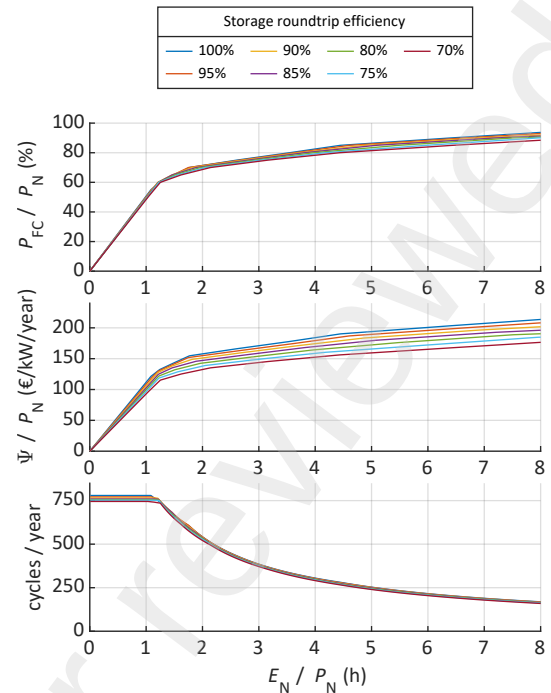


Figure 11: Impact of system efficiency on regulating power, annual profits, and cycles, for energy management through balancing market (strategy 2).

more power can be dedicated to providing FCR-N. In contrast, the profit is more sensitive to the storage efficiency, as seen in the second subplot. For a storage efficiency of 70 %, the profit is on average 14.9 % lower compared to a system with a roundtrip efficiency of 95 %. The lower profit is a result of both reduced capacity payment and higher energy costs for recovering the SOC. The annual number of full cycles appears to be relatively robust to the system efficiency. Until an E_N/P_N ratio of 1 h, the number of cycles is constant, because the maximum droop power P_{FC} increases linearly. Once the increase of P_{FC} stagnates, the annual number of cycles follows a hyperbolic decline.

The results for the exemption agreement are summarized in Figure 12. As seen from the first subplot, P_{FC} shows a strong dependency on the efficiency. With a storage efficiency of 95 % the full power can be applied from an E_N/P_N ratio of 0.53 h. In comparison, with an efficiency of 70 % the full power can be applied from an E_N/P_N ratio of 4.7 h. The reduced maximum power values of low storage efficiencies are directly reflected in the profit, due to the lower capacity payment. Even after allowing the maximum power, systems with a lower efficiency still achieve less profits due to the higher overall energy consumption. The high divergence in the maximum droop power is also reflected in the number of cycles. Systems with a high efficiency can dedicate more power to providing frequency control and will therefore experience a higher number of cycles. However, the cycle progressions for different efficiencies converge faster than the P_{FC} progressions. This is due to the fact that systems with a low efficiency have an overall higher energy consumption,

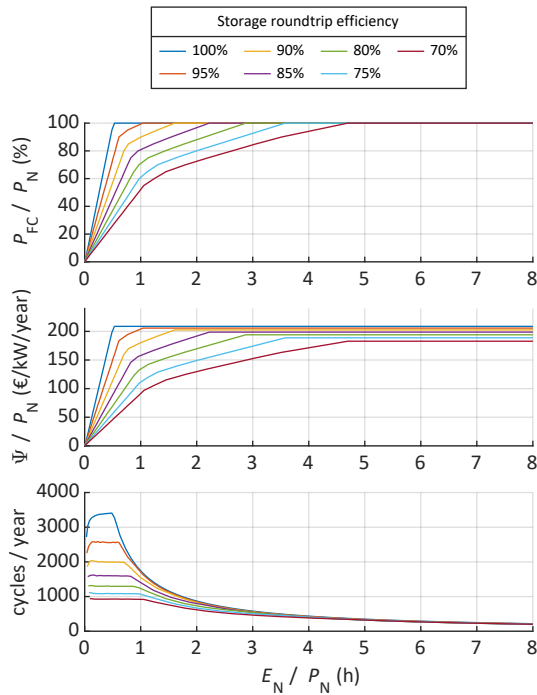


Figure 12: Impact of system efficiency on regulating power, annual profits, and cycles, for exemption agreement with Energinet (strategy 3).

and consequently, a higher energy throughput when following the same power profile.

The obtained results indicate that the operation with exemption agreement is more sensitive to the system efficiency. In this regard, it must be considered that batteries are only allowed to restore their reserves after delivering the service equivalent to full power in 15 min. This energy equivalent is estimated at the grid connection point and, thus, does not take into account the system efficiency. If the condition for energy recovery is not triggered for a longer period, systems with a low efficiency will experience a decline in their charge level. To avoid hitting the lower operational limit, these systems must dedicate less power to frequency control compared to more efficient systems, as shown by the results in Figure 12. This effect could be partly compensated by introducing a flexible reference power also for the operation with exemption. During longer periods without SOC restoration trigger, this can help to mitigate the effect of continuous depletion of the stored energy. However, from the exemption document it is not clear, whether this kind of operation is permitted.

5. Conclusions

Batteries are crucial system components in highly renewable power systems, particularly for providing frequency control services such as FCR-N. Battery operators that offer this service in the Nordic synchronous area are required to 100 % fulfill the traded capacity and can be excluded from further market participation if they repeatedly fail to do so.

Hence, it is crucial to know which ratio of the available power capacity can be allocated to delivering the service without violating the energy limits of the storage system. Focused on the Nordic power system with three years of frequency, market and tariff data, the present study addresses this issue and compares different energy recovery strategies for managing the energy level of batteries while providing primary frequency control. For each of the strategies, the potential annual profit was determined as a function of the energy/power capacity ratio and the efficiency of the storage system. It was found that for all strategies, capacity payment is the strongest factor that determines the overall profit. The highest regulating power, thus profit, can be achieved by signing an exemption agreement with the Danish TSO Energinet, which allows batteries to restore their energy level after an energy equivalent of full power provision for 15 minutes. In Sweden, Norway, and Finland, where this agreement is not available, the energy level can be managed by regulating the battery reference power on an hourly basis. The results clearly suggest that the energy associated with the reference power adjustment should be settled through the balancing market. Trading the energy one hour ahead on the intraday market slightly reduces the costs of energy, however, it allows significantly lower regulating power and therefore yields less overall profit. All in all, the presented data guides battery operators to: 1) Decide for a suitable operating strategy for FCR-N, based on system properties such as energy and power capacity, as well as roundtrip efficiency, 2) select the regulating power that achieves the highest profit without violating the service agreement, 3) know the annual cycles the battery will undergo when delivering FCR-N.

Future studies could apply the same methods for other services such as frequency containment reserve for disturbances (FCR-D) or fast frequency reserve (FFR). Furthermore, the presented results on expected annual profit and battery cycles can be used to determine optional energy/power capacity ratios for storage systems that promise the highest lifetime value. This can be done for various storage technologies, taking into account their respective characteristics such as average efficiency, cycle life, and investment costs.

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